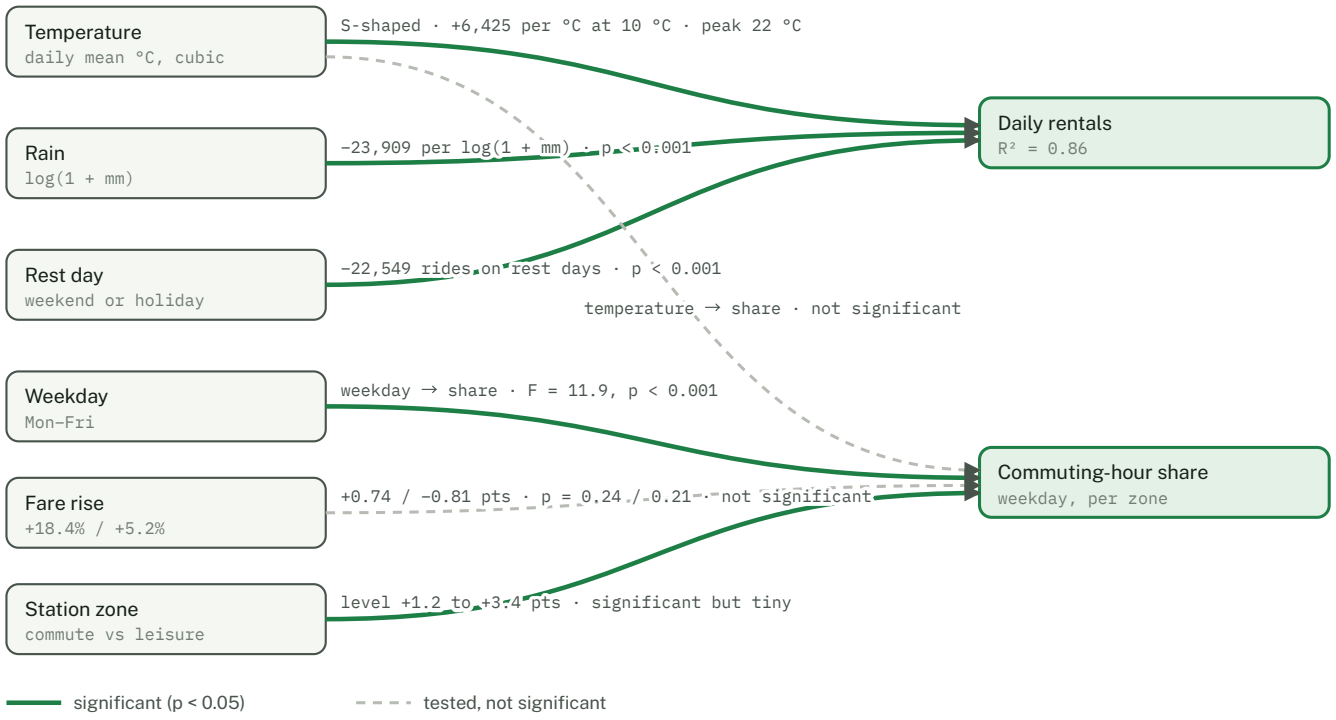
Three System Maps Comparison Report

What drives 따릉이 use, drawn three ways: from our regression results (Map A), from theory including features our data cannot see (Map B), and from a machine-learning model (Map C). The justification section explains why variables that were not significant can still matter, and what policy would miss by ignoring them.

Original hypotheses. Q1: daily rentals rise with temperature (H_0: $\beta_1 = 0$). Q2: after a fare rise, the weekday commuting-hour share at commute-zone stations changes (two-sided, $\alpha = 0.05$; meaningful if ≥ 3 points). Q3: the leisure zone does not change. Premise tests P1–P3: rest-day use, park-station share, weather sensitivity of commuting-hour rides.

Map A · Statistical significance (from the regression results)



Solid green: significant at p < 0.05. Dashed grey: tested and not significant. Only relationships our regressions could estimate appear.

Daily rentals, 2022 (main model, HAC standard errors)

Term	coef.	SE	p
Temperature (centred)	5,582.2	297.4	< 0.001
Temperature^2	-179.1	14.1	< 0.001
Temperature^3	-10.4	1.1	< 0.001
log(1 + rain mm)	-23,909.0	1,477.4	< 0.001
Rest day	-22,549.3	2,729.7	< 0.001

Reading the main model

R^2 0.861; largest VIF 7.05; Durbin–Watson 1.41, so standard errors are HAC (7 lags). Use peaks near 22 °C on a dry workday; one more degree adds 6,425 rides at 10 °C. A day with 10 mm of rain has about 57,331 fewer rides than a dry day; a rest day about 22,549 fewer.

Commuting-hour share: fare effect with and without controls (HAC standard errors)

Rise	Stations	Model	fare effect (pts)	95% CI	p
2023	commute zone	simple: share ~ post	+0.79	−0.62 to +2.20	0.27
2023	commute zone	multiple: + temperature + log rain + weekday	+0.74	−0.48 to +1.96	0.24
2023	commute zone	multiple: + temperature^2	+0.82	−0.43 to +2.07	0.20
2023	commute vs leisure	DiD + temperature + log rain + weekday	+0.22	−0.34 to +0.79	0.44

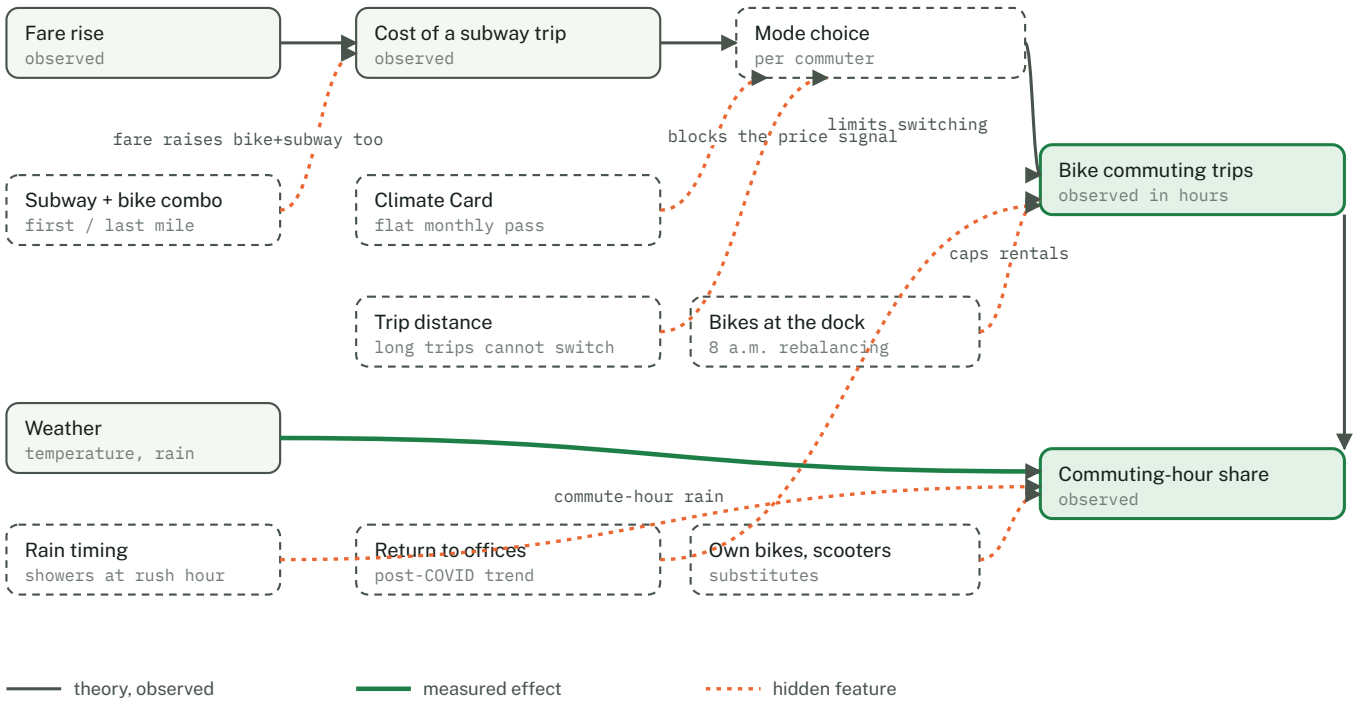
2025	commute zone	simple: share ~ post	−0.81	−2.06 to +0.45	0.21
2025	commute zone	multiple: + temperature + log rain + weekday	−0.81	−2.09 to +0.47	0.21
2025	commute zone	multiple: + temperature^2	−0.88	−2.15 to +0.39	0.17
2025	commute vs leisure	DiD + temperature + log rain + weekday	−0.19	−0.75 to +0.37	0.51

How we got to the main model (professor's feedback)

Model	Why this step	R^2	held-out R^2	AIC
1 simple	Week 4 plan: temperature only	0.458	0.451	8,736
2 + rain (mm)	Week 2 confounder: rain	0.666	0.656	8,561
3 + rest day	Week 2 confounder: day type (weekend/holiday)	0.722	0.714	8,496
4 + temperature^2	Brandon: the curve peaks in summer	0.780	0.771	8,413
5 log rain	the first millimetre of rain matters most	0.814	0.807	8,351
6 main: + temperature^3	**model 5 residuals were S-shaped across temperature**	**0.861**	**0.856**	**8,247**
7 + rain hours	hidden feature found by the ML map (section 17)	0.900	0.893	8,130
8 + month (sensitivity)	over-control check: month absorbs part of the temperature variation	0.890	0.874	8,185

Omitting rain biased the temperature slope down: 3,183 (simple) = 3,647 (with rain) + (-1,291 × 0.36); rain rises 0.36 mm per °C (r = 0.21).

Map B · Theory and scenarios, including hidden features

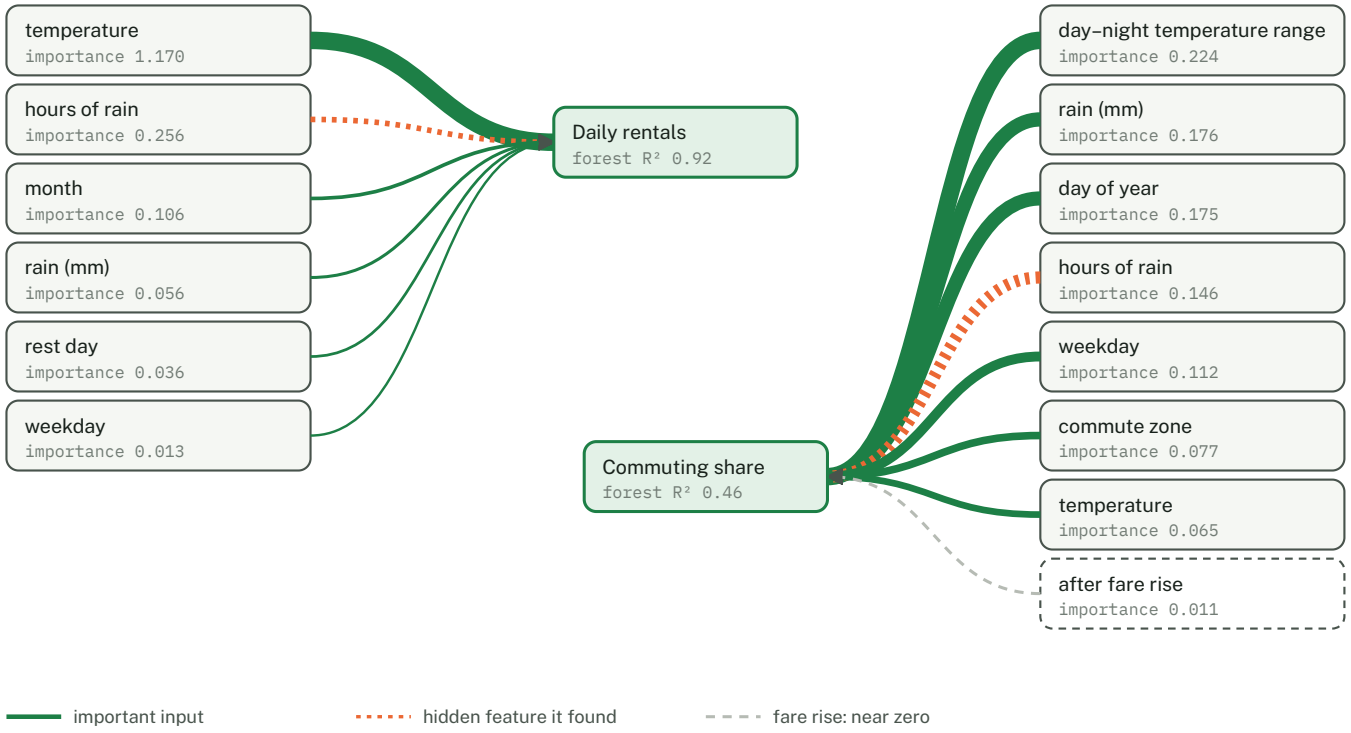


Dashed boxes: features our data cannot observe. Orange dotted arrows: their likely effect on the path from fare to commuting share.

Hidden feature	Mechanism	Effect on our estimate	How to measure it
Climate Card (flat pass)	Holders pay a fixed monthly price; a fare rise does not reach them	Pulls the 2025 effect toward zero	Card usage by line and hour
Bikes at the dock	Exit stations can empty at 8 a.m.; unmet demand never becomes a rental	Caps any rise in commuting rentals	Station availability snapshots at peak
Trip distance	Long commutes cannot switch to a bike at any price	Dilutes the average effect	Origin–destination trip data
Subway + bike combo	Bikes used for the first or last mile get more expensive with the subway	Could push the share down, not up	Rentals starting at exits within minutes of arrivals

Rain timing	A shower at 8 a.m. removes commuting rides; a night shower does not	Noise in the share that daily rain misses	Hourly precipitation
Return to offices	Post-COVID commuting recovery raises commuting rides everywhere	Inflates before–after estimates; removed by DiD	Office occupancy, subway ridership
Own bikes, scooters	Commuters who do switch may use their own bike or a shared scooter	Hides substitution from 따릉이 data	Bike counters, scooter operator data

Map C · Machine learning (random forest)



Arrow width = permutation importance (drop in held-out R^2 when the input is shuffled), averaged over 5 folds.

Held-out R^2, daily rentals

Model	R^2
simple regression	0.451
multiple regression (main)	0.856
random forest, same 3 inputs	0.841
random forest, all 8 inputs	0.918

Importance, daily rentals

Input	importance
temperature	1.170
hours of rain	0.256
month	0.106
rain (mm)	0.056
rest day	0.036
weekday	0.013
day–night temperature range	0.004
yesterday's rain	0.000

Held-out R^2, commuting share

Model	R^2
multiple regression	0.111
random forest	0.464

Importance, commuting share

Input	importance
day–night temperature range	0.224
rain (mm)	0.176
day of year	0.175
hours of rain	0.146
weekday	0.112
commute zone	0.077
temperature	0.065
after fare rise	0.011
leisure zone	0.010
summer window	0.001

With the same three inputs, our regression (held-out R^2 0.86) is as good as the forest (0.84). The forest pulls ahead only with extra inputs, mainly **hours of rain**; adding that one variable to the regression lifts R^2 to 0.90. For the share, the forest finds structure the linear model cannot (R^2 0.46 vs 0.11), all of it weather and calendar; the fare rise ranks near the bottom.

Comparing the maps

Question	Map A · significance	Map B · theory	Map C · ML
What drives daily rentals?	Temperature (curved), rain, rest days	Weather, calendar, supply	Temperature, then hours of rain
Does the fare move the commuting share?	No (n.s., both rises)	Should, unless hidden features block it	Near-zero importance
Does weather move the share?	No linear effect (n.s.)	Rain at commute hours should	Yes: rain and temperature range rank top
Main blind spot	Non-linear and unmeasured effects	Untested; needs new data	Correlation only; no causal claim

Justification document

Why "non-significant" variables might still matter

- **The fare rise.** Not significant in every design, yet the intervals still allow up to about 2 points (+2.00 in 2023). Across about 120,000 weekday rides a day in our windows, 1–2 points would be over a thousand commuting rides a day, enough to matter at single busy stations. A capped supply at exit docks and Climate Card holders could both be hiding part of it (Map B).
- **Weather in the share model.** Temperature and rain are not significant linearly (2023: p = 0.95 and 0.98), but the random forest ranks rain and the day–night temperature range at the top. The effect is non-linear and depends on timing, which a straight line cannot see.
- **Zone × fare.** The difference-in-differences is not significant, partly because name-based zones barely differ (1.2–3.4 points). A sharper, coordinate-based zone would test the mechanism with more power.
- **Heterogeneity.** An average of zero can hide opposite reactions: short-trip commuters switching to bikes while first/last-mile riders drop them because the whole trip got dearer.

Policy implications if we ignore the hidden features

- **Fare policy:** judging the 2025 rise without the Climate Card would understate how many riders felt no price change, and overstate how little commuters respond to price.
- **Bike supply:** planning on rentals alone treats empty docks as low demand. If peak demand is capped, adding bikes at exit stations could reveal a commuting use our data cannot see.
- **Demand forecasting:** forecasts built on daily rain totals miss commuting-hour showers; the ML map shows hours of rain carry more information than millimetres.
- **Evaluation:** before–after comparisons without a control group would credit the fare with the post-COVID return to offices (+0.66 points on average across random station groups in 2023).